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Next item →

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Almost there! Let's review your answers and then try again.

Your work showed a strong understanding of Q-learning and Markov processes, but there is room for growth in understanding neural network policies in reinforcement learning. Before retrying this assignment, we recommend focusing on this key area:

- **Deep Neural Networks with RL:** In Question 3, your understanding of neural network policies in reinforcement learning was incomplete. Review [this item](#) to better understand the role and characteristics of neural network policies, including their applicability to different action spaces and the necessity of learning state-action value functions.

Good job on the rest of the assignment!

Retry assignment



Instructions

1. What are the primary challenges in applying Q-learning to large Markov Decision Processes (MDPs) and the role of deep neural networks in approximating Q-values?

1 / 1 point

- ☒ Q-learning suffers from the curse of dimensionality in large MDPs.
- ☐ Deep neural networks eliminate the need for reinforcement learning algorithms.
- ☐ Q-learning can only be applied to discrete action spaces.
- ☐ Deep neural networks guarantee convergence in Q-learning tasks.

Correct
Correct! The exponential growth in the state-action space as the size of the MDP increases leads to an intractable number of Q-values that need to be computed and updated. This makes traditional Q-learning impractical for large MDPs due to both memory and computational constraints.

2. What is the role of the ϵ -greedy policy in Q-learning?

1 / 1 point

- ☐ To ensure the agent always selects the action with the highest Q-value.
- ☒ To facilitate exploration by occasionally choosing random actions while favoring high Q-value actions.
- ☐ To avoid all exploration by focusing solely on known state-action pairs.
- ☐ To reduce the computational complexity of Q-value estimation.

Correct
Correct. The ϵ -greedy policy promotes exploration by randomly selecting actions with probability ϵ , while exploiting high Q-value actions with probability $1 - \epsilon$.

3. Which of the following statements accurately describe the role and characteristics of neural network policies in reinforcement learning? Select all that apply.

0.4 / 1 point

- ☐ A neural network policy maps states directly to actions without requiring a value function.
- ☒ Neural network policies can be trained using policy gradient methods.

Correct
Correct! Policy gradient methods, such as REINFORCE or Proximal Policy Optimization (PPO), are commonly used to train neural network policies by optimizing the expected return with respect to the policy parameters. These methods adjust the parameters of the neural network to improve the policy's performance.

- ☐ Neural network policies are always more efficient than traditional methods.
- ☒ A neural network policy requires the state-action value function to be learned first.

This should not be selected
Incorrect. Neural network policies do not require the learning of a state-action value function (like Q-values) beforehand. Instead, they directly learn a policy that maps states to actions, which can be trained independently using policy gradients or other techniques.

- ☒ Neural network policies are only applicable to discrete action spaces.

This should not be selected
Incorrect. Neural network policies can be applied to both discrete and continuous action spaces. In continuous spaces, techniques like the Gaussian distribution are often used to parameterize the action space, allowing the network to output continuous values.

4. Which of the following statements correctly describes the concept of Markov processes and their memoryless property?

1 / 1 point

- ☒ Markov processes are stochastic processes where the future state depends only on the current state and not on the sequence of events that preceded it.
- ☐ Markov processes are deterministic processes with no randomness involved.
- ☐ Markov processes always converge to a single state regardless of the starting state.
- ☐ Markov processes require knowledge of all previous states to predict future states.

Correct
Correct! This is the definition of the memoryless property, which is a key characteristic of Markov processes. The future state is conditionally independent of past states, given the current state, meaning only the present state influences the future, not the history of events.

5. Which of the following statements correctly describes the role of Bellman equations in reinforcement learning?

1 / 1 point

- ☒ Bellman equations decompose the value function into immediate reward and discounted future values.
- ☐ Bellman equations provide a way to account for exploration in unknown environments.
- ☐ Bellman equations are used to model the transition probabilities in MDPs.
- ☐ Bellman equations help in dynamically adjusting the learning rate in Q-learning.

Correct
Correct! Bellman equations provide a recursive breakdown of the value function, separating the immediate reward from the expected future rewards, discounted by a factor. This decomposition is crucial in reinforcement learning, as it helps in calculating the optimal policy by iteratively updating the value function based on current and future rewards.